

Suraj Mundhada

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[LinkedIn](#) | [GitHub](#)

EDUCATION

Vishwakarma Institute of Technology (VIT)

B.Tech in Computer Engineering; CGPA: 8.86

Pune, India

Aug 2023 – May 2027

EXPERIENCE

Research Intern, Speech & NLP

Vishwakarma Institute of Technology (VIT) — Advised by Prof. Dr. Ganesh Bhutkar

Pune, India

Jul 2026 – Present

- Investigating **parameter-efficient adaptation (LoRA/PEFT)** of **OpenAI Whisper-small** for medical speech recognition, benchmarking against zero-shot and full encoder-decoder fine-tuning baselines on the MultiMed English corpus.
- Designing an evaluation pipeline on an unseen Indian-English clinical dataset (Eka Medical ASR), measuring **WER, CER, and medical-entity preservation** (medications, dosages, procedures) to evaluate cross-domain robustness.
- Building a downstream NLP module that converts ASR transcripts into **structured, EMR-ready documentation drafts** for physician review, with no autonomous clinical decisions.

TECHNICAL SKILLS

Languages: Java, Python, C++, SQL, JavaScript, Shell Scripting

Frameworks: Spring Boot, Spring Security, React, Next.js

AI/ML: LLMs, NLP, RAG, LangChain, OpenAI Whisper, Llama 3

Cloud & Tools: AWS, Docker, Git, GitHub, GitHub Actions, Linux, Maven

Databases: MySQL, MongoDB

Coursework: Data Structures & Algorithms (DSA), OOP, DBMS, Operating Systems (OS), Computer Networks, System Design

AI-Assisted Development: Cursor, GitHub Copilot, ChatGPT, Claude Code

PROJECTS

TrailBuddy: Full-Stack Travel Marketplace

Mar 2026 – Apr 2026

Technologies: Java, Spring Boot, MySQL, Spring Security, React, Docker, WebSockets

- Developed a full-stack marketplace with **Spring Boot REST APIs**, JWT authentication and role-based authorization for traveler, guide and booking workflows.
- Integrated **Razorpay Sandbox, Google Places API and Gemini API**, and implemented **WebSocket-based real-time messaging** between users and travel guides.
- Containerized the application using **Docker** with MySQL-backed persistent workflows; recognized as a **Top 10 Finalist** at the MVPMM Startup Hackathon 2026.

CliniQ+: Real-Time Clinical AI & RAG Assistant

Jan 2026 – Mar 2026

Technologies: Next.js, React, TypeScript, MongoDB, Deepgram, Llama 3, LangChain, Qdrant

- Built a real-time **speech-to-EMR pipeline** using **Deepgram WebSocket APIs**, converting doctor-patient conversations into structured clinical transcripts.
- Built a **RAG pipeline** using **LangChain, BGE embeddings and Qdrant** for semantic retrieval, integrating retrieved context with **Llama 3 via Groq** to generate structured EMR fields.
- Reduced the proportion of consultation time spent on documentation from **45% to 20%** by automating speech transcription and **LLM-based EMR extraction**.

RESEARCH PUBLICATIONS & PATENTS

Detection of Diabetic Retinopathy Using Python | ICTCS 2024 (Springer)

A Machine Learning Pipeline for Checking Junk Food Ingredients | IEEE GCWCN 2025

Granted South African Patent | ML Pipeline for Food Ingredient Analysis

ACHIEVEMENTS & LEADERSHIP

Flipkart GRID 8.0 National Semi-Finalist | Among 1,60,000+ Participants

Internal Smart India Hackathon 2025 Winner | Team QueueT Pi

3× Hackathon Finalist | Cavista Hackathon, MVPMM Heera Software Startup Hackathon, Symbiosis Skill Hackathon

Start-up Connect Coordinator, EDC | **Social Activity Coordinator, VIT SWD**

- Coordinated events for 1,000+ participants and led a 20+ member team; managed PRISM'24 operations (500+ attendees).